# Development of a Digital Twin Framework for Smart Farming: Coupling Computational Fluid Dynamics with Reduced-Order Microclimate Models to Predict Greenhouse Energy Balance



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# Outline

1. Background and significance
2. Literature review
3. Objectives
4. Scope of work
5. Expected Result


Google Earth
Data SIO, NOAA, U.S. Navy

# Background and Significance

## Greenhouse






**Location: U Thong Banana Garden**
10 Moo 14, Chorakhe Sam Phan, U Thong,
**SuphanBuri, Thailand**

**Crop : Cherry Tomato**

# Background and Significance

## Greenhouse





### Strength

- Prevent rain impact
- Protect plant from disease
- Virus control
- Radiation quality, diffused light and avoid direct light

### Risk

- If not enough ventilation it may cause Temperature buildUp .

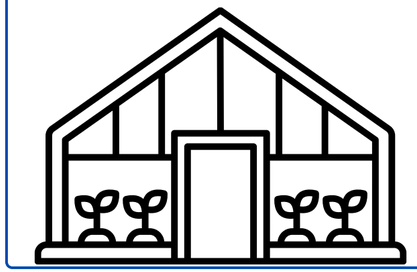
# Background and Significance

## Microclimate

distinct climate envelope within a small enclosed area (canopy layer)
where local heat, radiation, and moisture differ sharply from outdoor weather.

**Key Parameters**

- Temperature
- Relative humidity
- Solar gain
- Ventilation
- Transpiration
- Conduction
- Energy balance
- Mass balance





# Background and Significance

## The Bottleneck



Greenhouse

Design : Empirical Design

IoT : Isolated & Delay

CFD : Too Slow

## The Framework

CFD

Reduced order modeling

Energy Balance Model

Onsite IoT

## The Impact

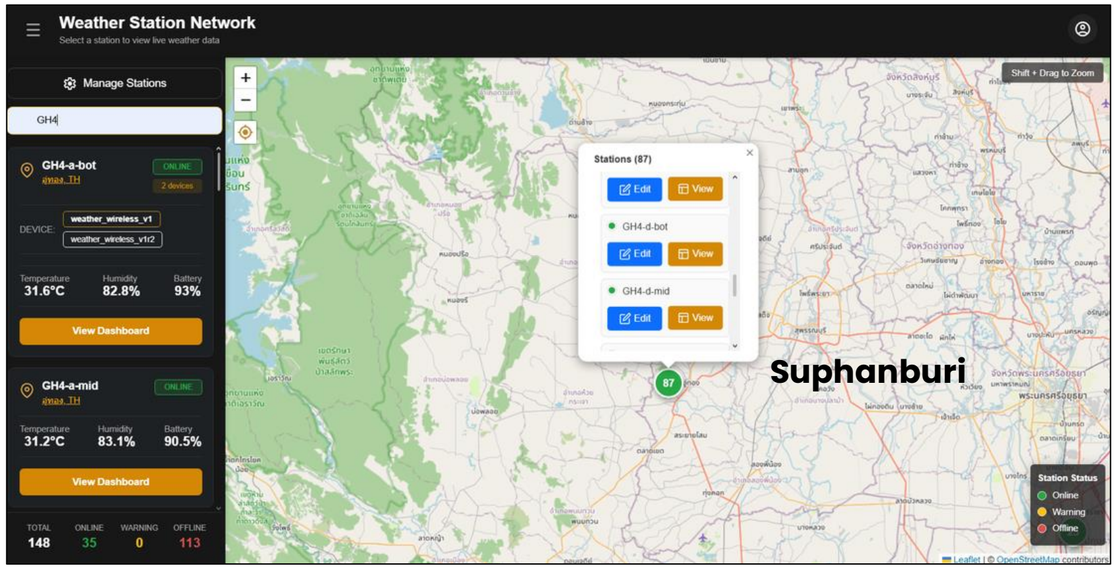


Real-time Predictive Microclimate Model



# Background and Significance

## Internet of Things (IoT)







**4 Poles**

**3 Sensors each Pole**

**Total 12 sensors**

Cover High and Low spot







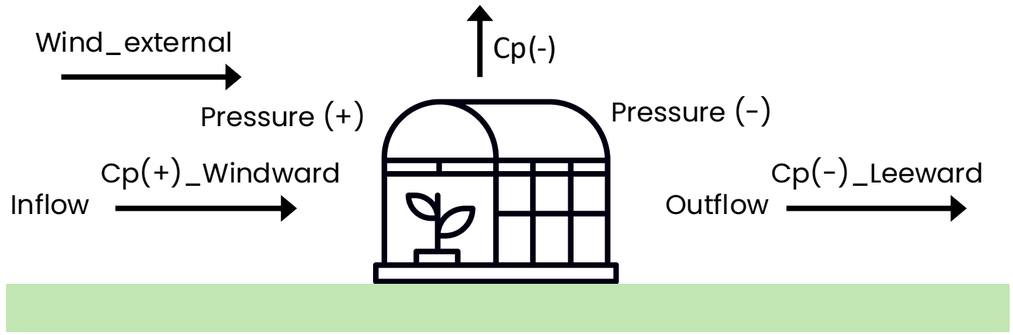
# Related Theory

# Related Theory

## Natural ventilation

### Wind-Driven Ventilation

- Airflow and pressure behavior around the greenhouse




**Windward Side**

- Positive pressure
- Pressure Coefficient (+)
- Natural air inlet opening

**Leeward Side**

- Negative pressure
- Pressure Coefficient (-)
- Natural air outlet opening

### Stack Effect / Buoyancy-Driven

- Ideal Gas Law:

$$P = \rho \cdot R \cdot T$$

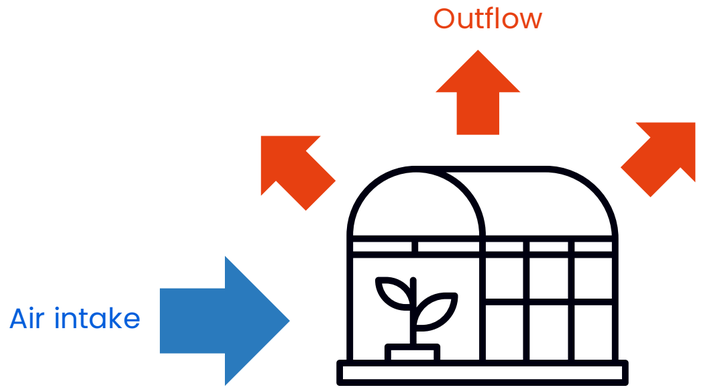
P: Absolute pressure (Pa)

ρ: Density ($kg/m^3$)

R: Specific gas constant (J/ (kg· K))

(Dry air: R = 287 J/ (kg· K))

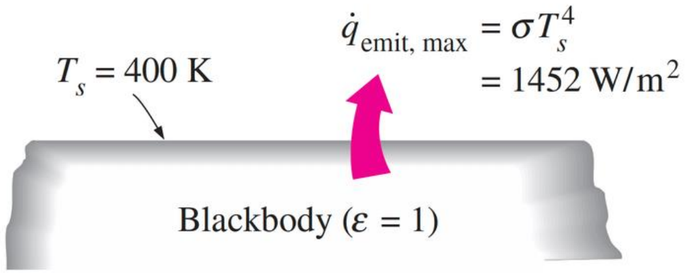
T: Temperature (K)



# Related Theory

## Radiation

### Emission

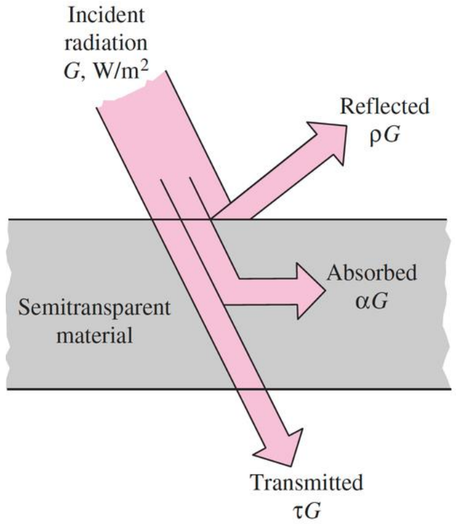



$$E = \sigma T^4$$

Shortwave (high Temperature)

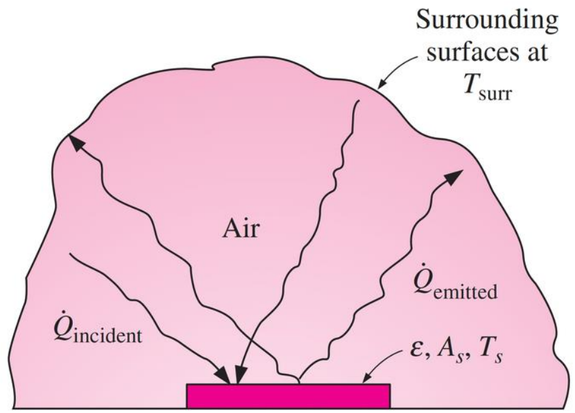
Longwave (low Temperature)

### At a Surface




$$\rho + \tau + \alpha = 1$$

### Between Two Surfaces




$$Q = \varepsilon_1 \varepsilon_2 F A \sigma (T_1^4 - T_2^4)$$

Source: Yunus A. Cengel, Heat Transfer: A Practical Approach, 2004

# Related Theory

## Reduced-Order Model (ROM)

### 1. What is ROM (Reduced-Order Model)?

computational simplification technique that captures the dominant behavior of a complex, high-dimensional system using a drastically smaller set of degrees of freedom.

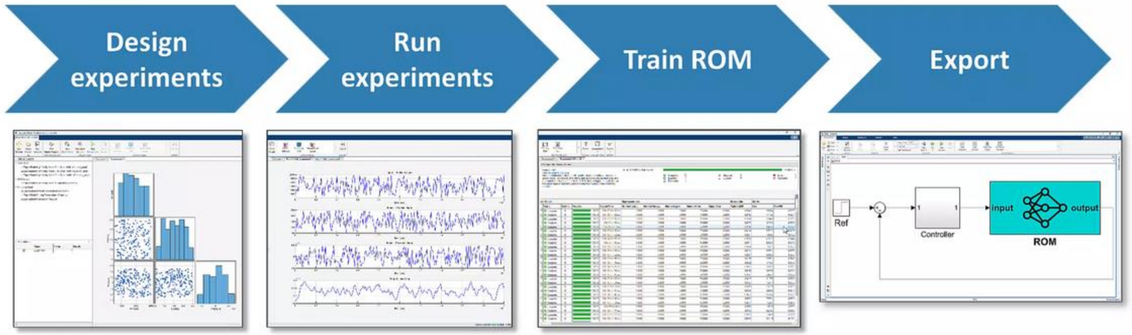
### 2. What is POD (Proper Orthogonal Decomposition)?

a mathematical technique to extract a set of data snapshots to orthonormal basis from .

| Component | Role | Function |
|---|---|---|
| POD | Feature Extractor | Finds what spatial patterns matter most from simulation snapshots. |
| ROM | System Model | Uses POD modes to predict system response over time/parameters at fraction of the cost. |

### CFD to ROM

1. Collect Snapshots CFD 30-50 Case
2. Write python to combine data into Matrix
3. Extract “Modes” with SVD in Python
4. Matrix to ROM (A and B direction)




Source: G Berkooz, P Holmes, J L Lumley. 1993. The Proper Orthogonal Decomposition in the Analysis of Turbulent Flows. Annual Review of Fluid Mechanics.
Picture:https://blogs.mathworks.com/deep-learning/2024/10/21/introducing-the-reduced-order-modeler-app-how-to-interactively-create-ai-based-reduced-order-models-roms/

# Literature review

# Literature review

**Authors/year:** Samson Sogbaike, Ibrahim Albayati, Amir Badiee, Chris Bingham, Aliyu Aliyu, 2026
**Article:** Transient CFD analysis of greenhouse microclimate under temperature-regulated heating

## Set up Model

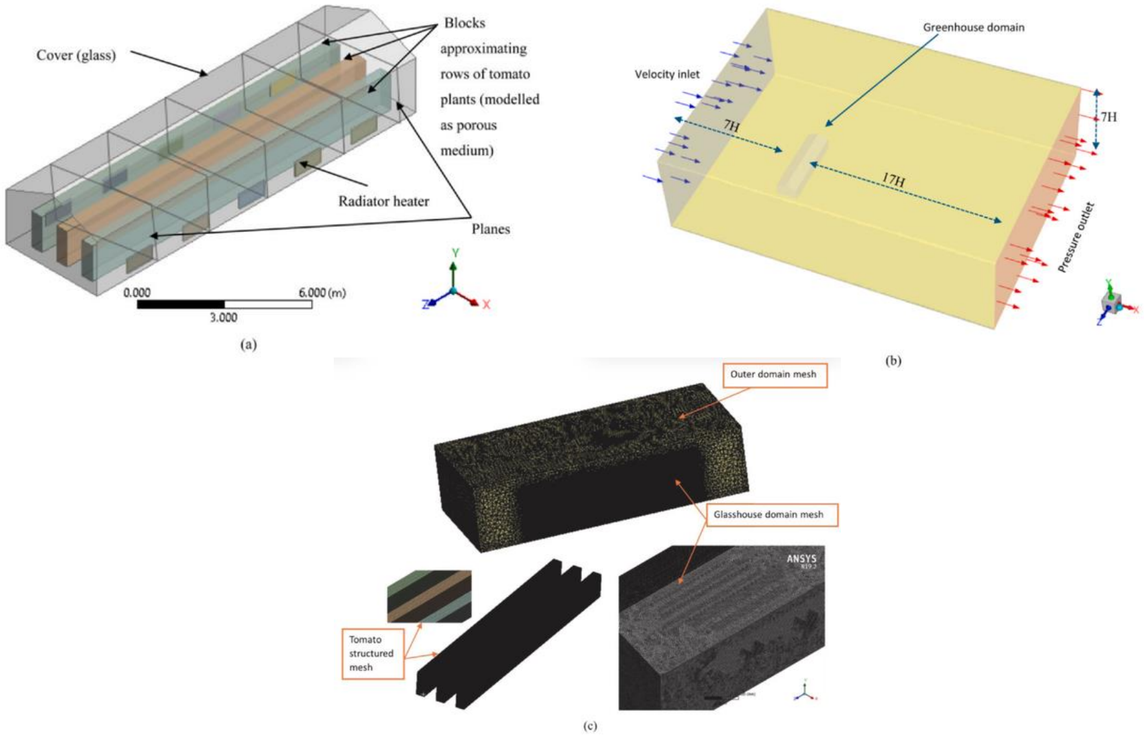



Fig. 1. (a) Greenhouse geometry with tomato crop, radiator heater and planes (represented as lines) along the z-axis for data extraction (greenhouse facing the west, and the 3D-coordinate represents x as east, y as zenith, and z as south), (b) Computational domain of the greenhouse surrounded by an outdoor region (air/fluid domain), (c) section view of the unstructured mesh of the outer domain with greenhouse domain mesh, structured mesh of the tomato crop canopy and unstructured mesh of the greenhouse domain.

**Authors/year:** Jörg Franke, Antti Hellsten, Heinke Schlünzen, Bertrand Carissimo, 2007
**Article:** Best Practice Guideline for the CFD Simulation of Flows in the Urban Environment

## Set up of Air domain

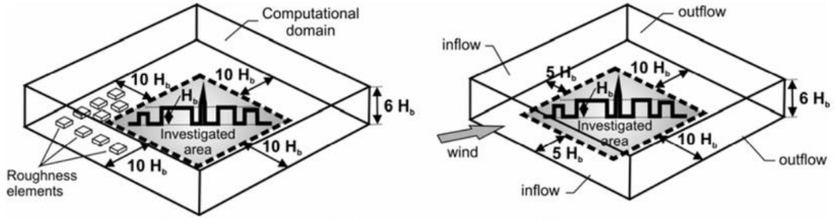
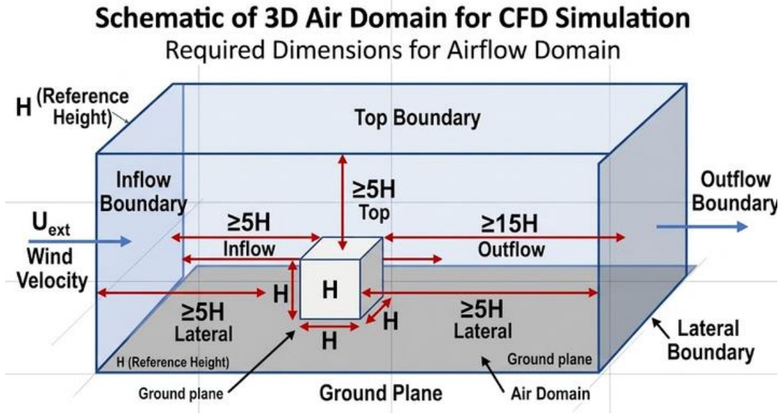



**Figure 1: MISKAM boundaries in multidirectional and one directional simulation if the roughness around the investigated area is homogeneous**

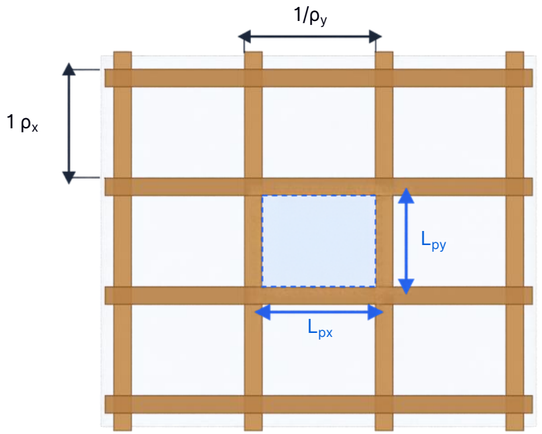
## Size of Air domain




Source: Samson Sogbaike, Transient CFD analysis of greenhouse microclimate under temperature-regulated heating, 2026
Jörg Franke, Antti Hellsten, Best Practice Guideline for the CFD Simulation of Flows in the Urban Environment, 2007
AI Generate by GEMINI

# Literature review

**Authors/year:** D. L. Valera, A. J. Álvarez and F. D. Molina, 2006
**Article:** Aerodynamic analysis of several **insect-proof screens** used in greenhouses






- **The average length of the pores in the two main directions of the mesh**

$$L_{px} = \frac{1}{\rho_y} - D_h \; ; \; L_{py} = \frac{1}{\rho_x} - D_h \qquad [1]$$

- **The porosity** $\alpha$ ($m^2\ m^{-2}$) expresses the relationship between the surface area of the pores ($S_p$), and that of the total mesh ($S_t$)

$$\alpha = \frac{S_p}{S_t} = \frac{L_{px}\ L_{py}}{(L_{px} + D_h)\ (L_{py} + D_h)} \qquad [2]$$

Mesh thread

Open pore (used to measure $L_{px}$, $L_{py}$)

**Thread diameter ($D_h$)**




Source: D. L. Valera, A. J. Álvarez and F. D. Molina, Aerodynamic analysis of several insect-proof screens used in greenhouses, 2006

**Authors/year:** António F. Miguel, 1998
**Article:** Airflow through **porous screens**: from theory to practical considerations

- **The equation coefficients** that best fitted the data obtained were found through a logarithmic regression analysis.

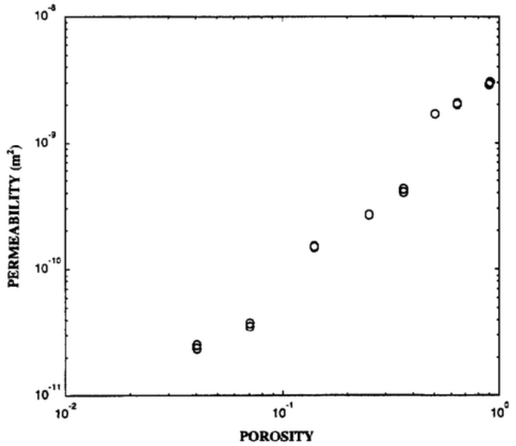



Fig. 7. Screen permeability $K$ as a function of porosity $\varepsilon$ for all the screen samples tested.

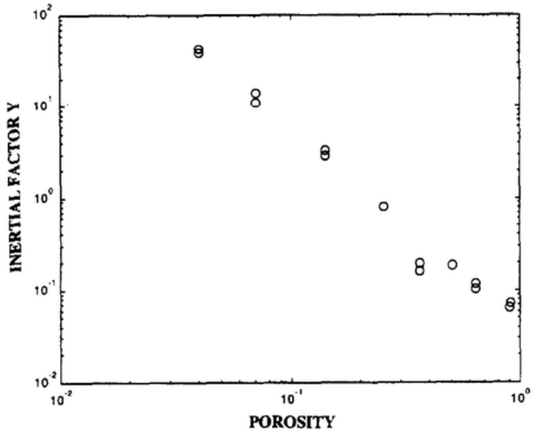



Fig. 8. Inertial factor $Y$ as a function of porosity $\varepsilon$ for all the screen samples tested.

$$K = 3.44 \times 10^{-9} \varepsilon^{1.6}$$

$$Y = 4.30 \times 10^{-2} / \varepsilon^{2.13}$$

Source: António F. Miguel, Airflow through porous screens: from theory to practical considerations, 1998

# Literature review

**Authors/year:** Yu-Qing Tang, Wen-Zhen Fang, Chun-Yu Zheng, Wen-Quan Tao, 2025
**Article:** Applications of **POD-based reduced order model** to the rapid prediction of velocity and temperature in data centers

## Data center Model

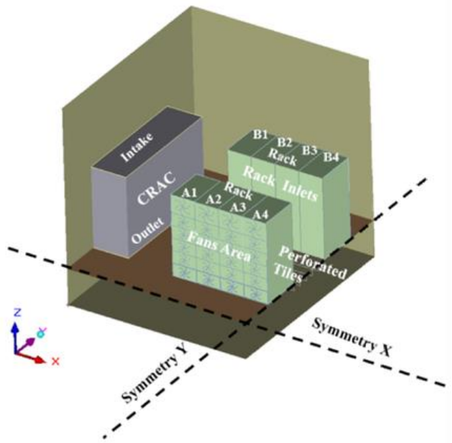



Fig. 1. (a) Computational domain for one quarter of the data center

## POD temperature contour

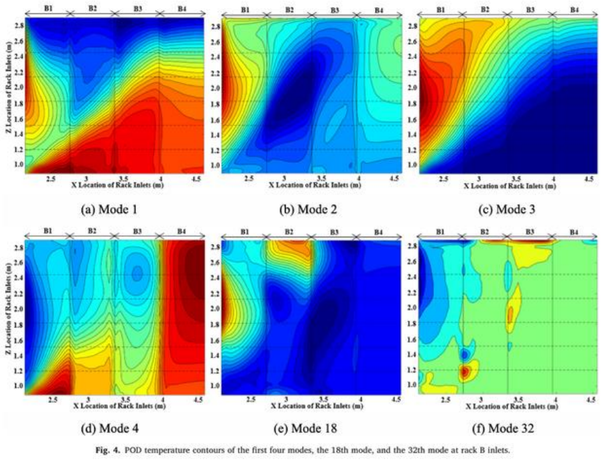



Fig. 4. POD temperature contours of the first four modes, the 18th mode, and the 32th mode at rack B inlets.

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## Result

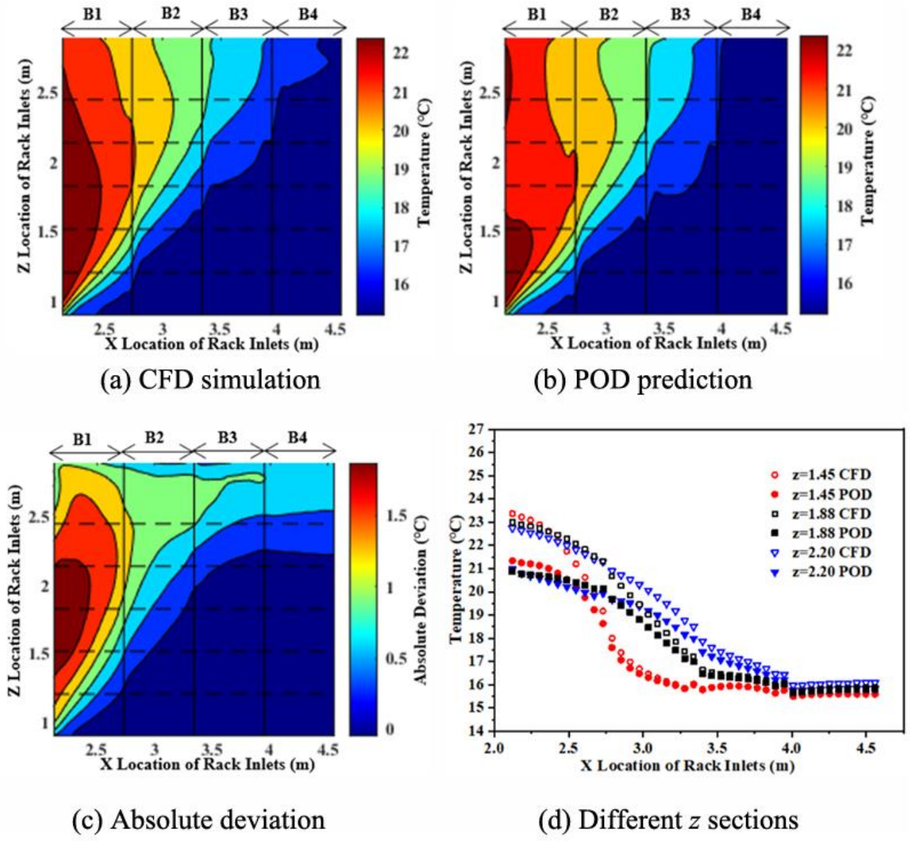



Fig. 7. Temperature contours of B1-B4 rack inlets at predicted case and comparison between CFD and POD results at different $z$ locations.

Source: Yu-Qing Tang, Applications of POD-based reduced order model to the rapid prediction of velocity and temperature in datacenters, 2025

# Literature review

**Authors/year:** Samuel Sourisseau, Cyril Toublanc, Etienne Chantoiseau, Michel Havet, 2026
**Article:** Contributions to the development and simulations of generic, modular and **multiphysics greenhouses dynamic models**, evaluated with a whole year study case dataset

## Experimental Greenhouse Description

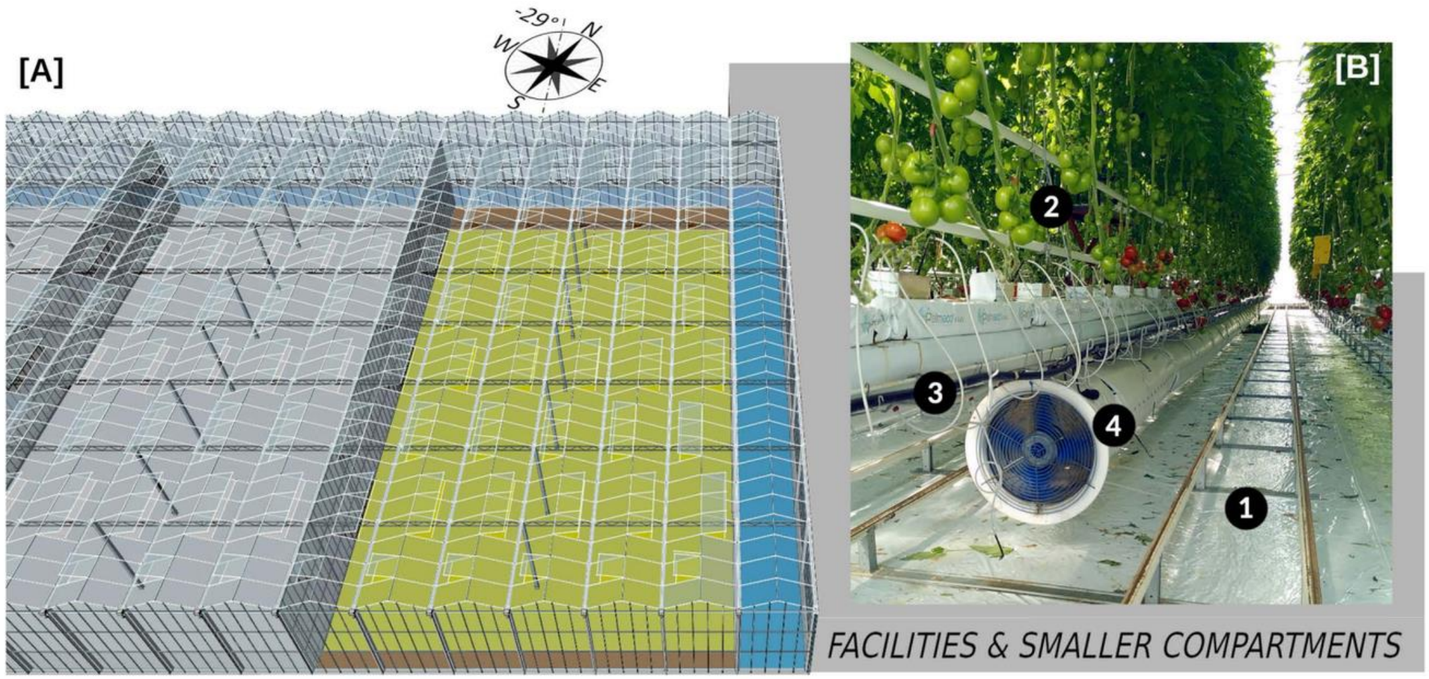



**Fig 1. A studied greenhouse compartment general drawing and a picture from inside.** (A) The studied greenhouse compartment (yellow) with its alleyways (brown), its adjoining corridors on the east and north sides (blue), and another similar compartment on its west side. (B) Picture taken in the compartment. In addition to three heating pipe networks (Rails 51 (1), Forcas (2), PE (3)), a perforated air mixing duct (4) is installed below 4 out of the 15 tomato rows.

## Models Development

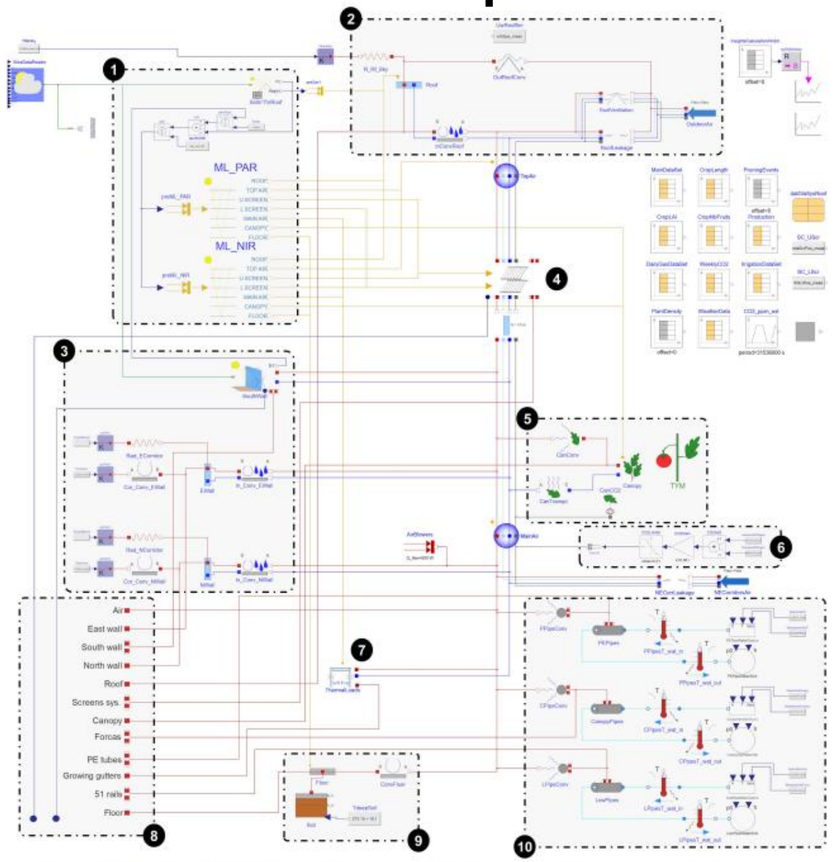



**Fig 4. Top-level graphical view of the global model of the greenhouse compartment.** Between the elements, connectors of different types are used. Red: heat transfer characterized by a temperature (K) and a heat flow (W) | Yellow: solar heat flux (W m$^{-2}$) | Light blue: dry airflow characterized by a partial pressure (Pa) and a mass flow (kg s$^{-1}$) | Dark blue: water vapor flow (same characterization) | Grey: $CO_2$ flow (same characterization) | Green: weather bus, for the meteorological data and the sun position | Turquoise: hot water flows.

**Source: Samuel Sourisseau, Cyril Toublanc, Etienne Chantoiseau, Michel Havet, Contributions to the development and simulations of generic, modular and multiphysics greenhouses dynamic models, evaluated with a whole year study case dataset, 2026**

# Literature review

**Authors/year:** Samuel Sourisseau, Cyril Toublanc, Etienne Chantoiseau, Michel Havet, 2026
**Article:** Contributions to the development and simulations of generic, modular and **multiphysics greenhouses dynamic models**, evaluated with a whole year study case dataset

## Result

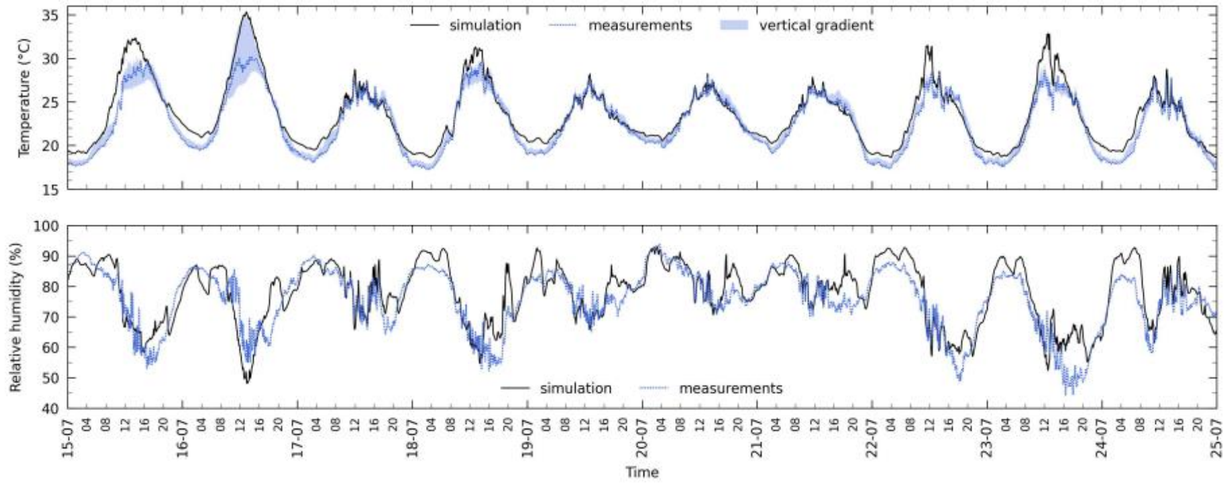



**Fig 9. Temperature and relative humidity in the MainAir zone during 10 summer days.** Main air zone temperature (top) and relative humidity (bottom) during 10 summer days from the 15th to the 25th of July 2015 (black solid line: simulation | blue dotted line: "process" measurements $T_{air,in}$ and $RH_{air,in}$ (at the plant apex)). The blue area on the upper graph delimits the temperature vertical gradient as the difference between the minimum and maximum of $T_{air,in,L}$, $T_{air,in,M}$, $T_{air,in}$ and $T_{air,in,H}$.

## Result

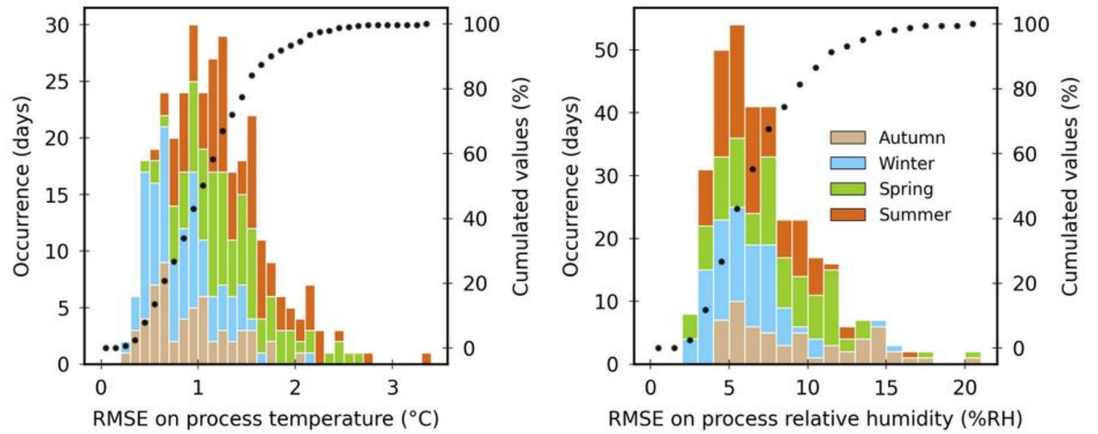



**Fig 10. Distributions of the daily RMSE based on the process temperature (left) and relative humidity (right).** Bars are colored according to the season of the day. The dotted curves correspond to the cumulated values (right axis of each figure).

**Source: Samuel Sourisseau, Cyril Toublanc, Etienne Chantoiseau, Michel Havet, Contributions to the development and simulations of generic, modular and multiphysics greenhouses dynamic models, evaluated with a whole year study case dataset, 2026**

# Objectives

- To develop **3D Computational Fluid Dynamics (CFD)** for internal microclimate airflow velocity patterns, and thermal distribution.
- To **integrate** and couple the **CFD-based Reduced-Order Model with the energy balance** framework into a **Digital Twin Framework** for real-time smart farming applications.

# Scope of work




**Greenhouse**
**&**
**Weather Station**

Ansys




**Computational**
**Fluid Dynamics (CFD)**

python™

**Reduced Order Modeling (ROM)**

OpenModelica



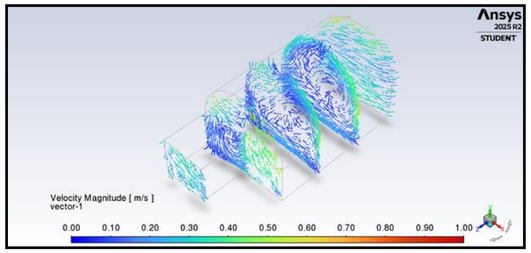

**OpenModelica**

python™ + OpenModelica

**Digital Twin Framework**

# Expected Results

Ansys




**Computational Fluid Dynamics (CFD)**

python™

**Reduced Order Modeling (ROM)**

OpenModelica

**OpenModelica Energy Balance**

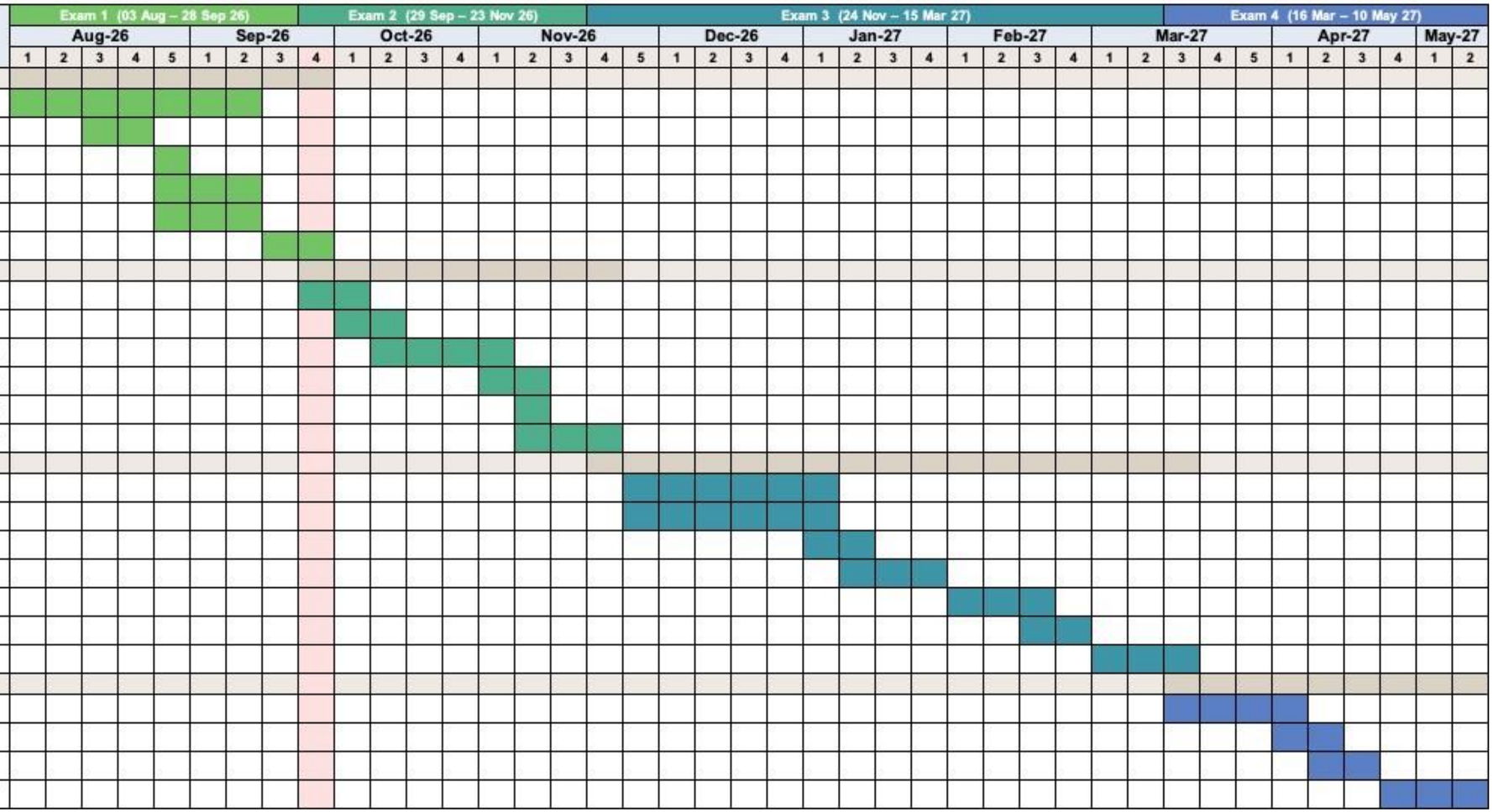


**Real greenhouse adaptation Digital Twin Framework**

# Gantt Chart

## Senior Project Schedule

| WBS | TASK | START | END |
|---|---|---|---|
| **Phase 1: Project I Midterm working plan (03 Aug 26 – 28 Sep 26)** | | | |
| 1.1 | Literature review | 03-Aug-26 | 18-Sep-26 |
| 1.2 | Survey and collecting data | 20-Aug-26 | 30-Aug-26 |
| 1.3 | 3D Modeling in SolidWorks | 31-Aug-26 | 01-Sep-26 |
| 1.4 | Software Training | 01-Sep-26 | 20-Sep-26 |
| 1.5 | Parameter Design | 04-Sep-26 | 20-Sep-26 |
| 1.6 | Presentation preparation | 21-Sep-26 | 28-Sep-26 |
| **Phase 2: Project II Final working plan (29 Sep 26 – 23 Nov 26)** | | | |
| 2.1 | Collecting data from ERA5 to reanalyse | 01-Oct-26 | 06-Oct-26 |
| 2.2 | Selecting data to validate | 07-Oct-26 | 12-Oct-26 |
| 2.3 | Parameter set up | 12-Oct-26 | 02-Nov-26 |
| 2.4 | Ansys Fluent and Modelica Simulation Trial | 03-Nov-26 | 09-Nov-26 |
| 2.5 | Discussion and conclusion | 09-Nov-26 | 13-Nov-26 |
| 2.6 | Presentation preparation | 14-Nov-26 | 23-Nov-26 |
| **Phase 3: Project III Midterm working plan (24 Nov 26 – 15 Mar 27)** | | | |
| 3.1 | Ansys Fluent Simulation | 02-Dec-26 | 05-Jan-27 |
| 3.2 | Modelica Simulation | 02-Dec-26 | 05-Jan-27 |
| 3.3 | Validation between simulation and IoT data | 06-Jan-27 | 13-Jan-27 |
| 3.4 | ROM Extraction | 14-Jan-27 | 31-Jan-27 |
| 3.5 | Validation between ROM and IoT data | 01-Feb-27 | 15-Feb-27 |
| 3.6 | Discussion and conclusion | 16-Feb-27 | 28-Feb-27 |
| 3.7 | Presentation preparation | 01-Mar-27 | 15-Mar-27 |
| **Phase 4: Project IV Final working plan (16 Mar 27 – 10 May 27)** | | | |
| 4.1 | Integration between Modelica and ROM | 17-Mar-27 | 09-Apr-27 |
| 4.2 | Validation with IoT data | 10-Apr-27 | 17-Apr-27 |
| 4.3 | Discussion and conclusion all phase | 18-Apr-27 | 25-Apr-27 |
| 4.4 | Presentation preparation | 26-Apr-27 | 10-May-27 |



# Q&A

# Backup

# Additional

## POD-ROM

1. What is ROM (Reduced-Order Model)?
   - It's same Caricature for reduce variables
   - Computer calculate within a short period of time

2. What is POD (Proper Orthogonal Decomposition)?
   - Keep what is important
   - Find "Modes" (%)

**Step CFD to ROM**

1. Run CFD 30-50 Case
2. Write python to combine data into Matrix
3. Extract "Modes" with SVD in Python
4. Matrix to ROM (A and B direction)

**A.Non-Intrusive ROM** [know Temp, Velocity inlet, coefficient] created predictive function (POD insert-ML)

**B.Intrusive ROM (Projection-based / POD-Galerkin)** projected "phi" on Navier-Stokes equation [PDE to ODEs] then solve ODEs

| Comparision | CFD | POD-ROM |
|---|---|---|
| Calculation time | 2-8 hr/case | 0.01-5 sec |
| Viewpoint | 3D image/Slow | 3D image/Fast |

# Additional

## SVD (Singular Value Decomposition)

$$S = U \sum V^T$$

| Component | Mathematical Name | Meaning in CFD / ROM | Simple meaning |
|---|---|---|---|
| U | Left Singular Vectors | Spatial Modes (Phi) | "What does the pattern look like?"<br><br>(e.g., where it's hot, wind direction) |
| Sigma | Singular Values | Energy Level / Importance | "How important is each pattern?"<br><br>(Sorted from most to least important) |
| $V^T$ | Right Singular Vectors | Temporal Dynamics | "How does the pattern change over time?"<br><br>(e.g., active at noon, drops at night) |

# Additional

## Porous-jump

**The average length of the pores in the two main directions of the mesh** $L_{px}$ and $L_{py}$ are the average length (in m) of the pores in the two main directions; $D_h$ is the average diameter (in m) of the threads which make up the mesh; $\rho_x$ and $\rho_y$ represent the number of threads per unit of length (threads $m^{-1}$) in each of the two main directions.

$$L_{px} = \frac{1}{\rho_y} - D_h \; ; \; L_{py} = \frac{1}{\rho_x} - D_h \qquad [1]$$

Data on permeability and on the inertial factor Y vs. porosity are presented in Figs. 7 and 8. The equation coefficients that best fitted the data obtained were found through a logarithmic regression analysis. These equations are

$$K = 3.44 \times 10^{-9} \varepsilon^{1.6} \qquad [8]$$

and

$$Y = 4.30 \times 10^{-2} / \varepsilon^{2.13} \qquad [9]$$

valid for $0.04 \leq \varepsilon \leq 0.90$.

Source: D. L. Valera, A. J. Álvarez and F. D. Molina, Aerodynamic analysis of several insect-proof screens used in greenhouses, 2006

Source: António F. Miguel, Airflow through porous screens: from theory to practical considerations, 1998